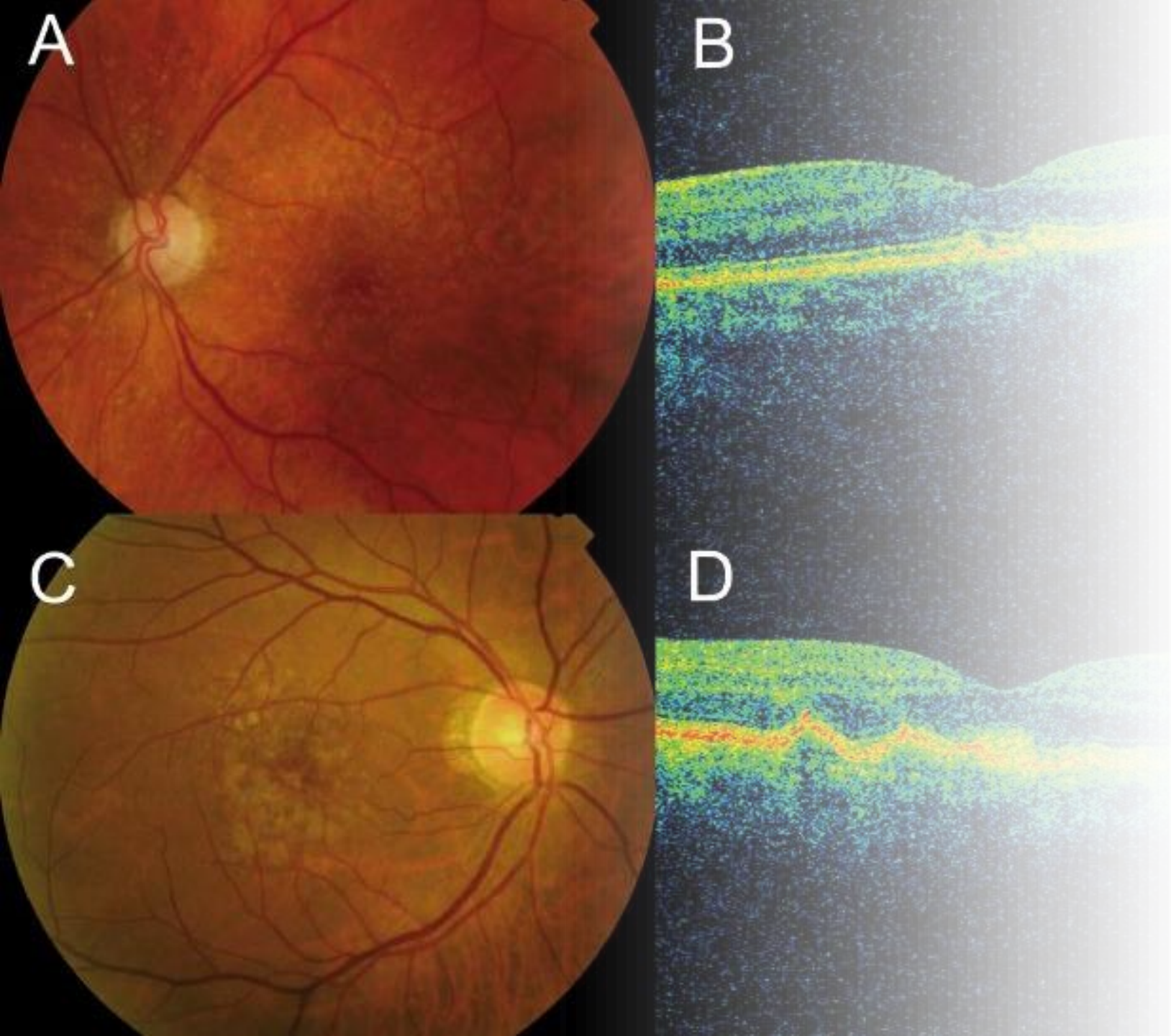
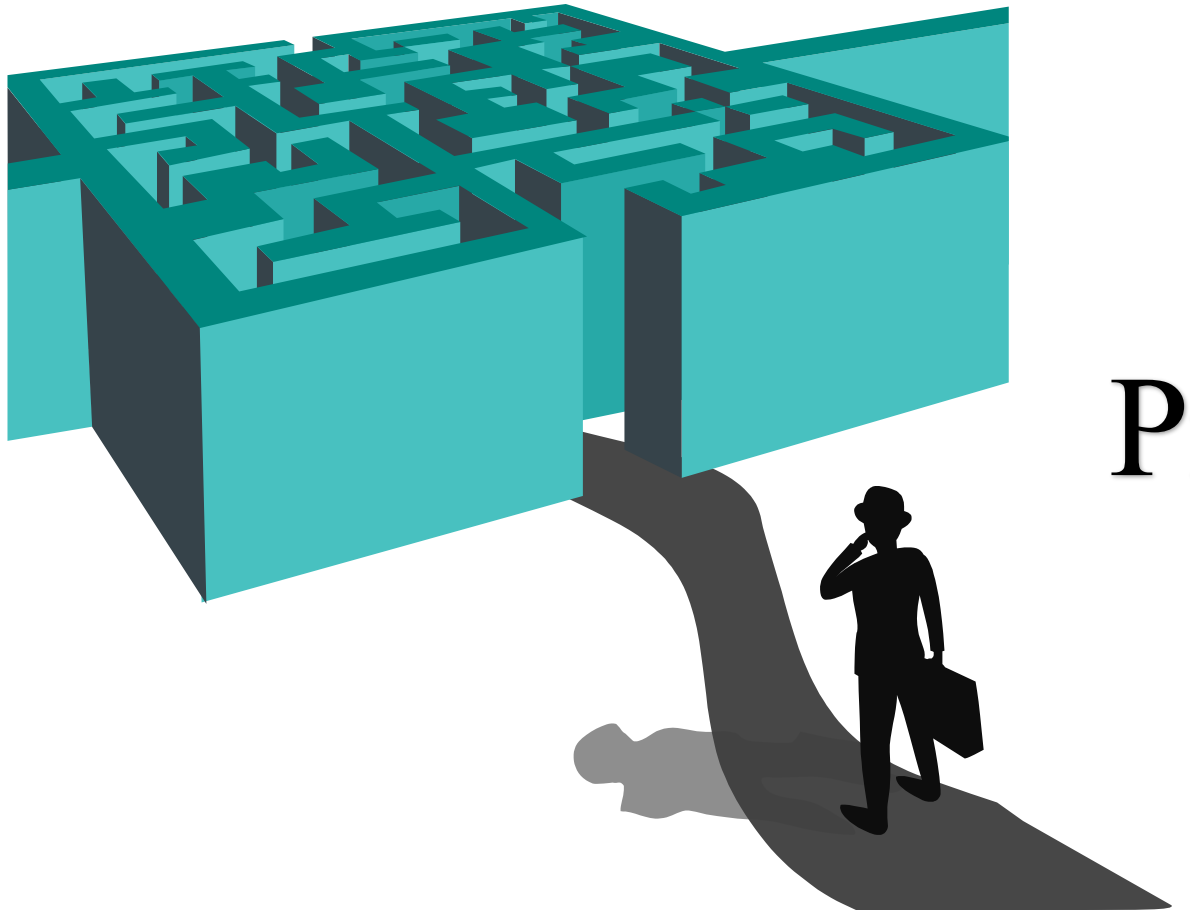


OCT Reconstruction and Volume Processing Pipeline

Presentation Overview

- Problem Statement
- 2D Processing (B-scan Level)
- Volume Processing



Problem Statement

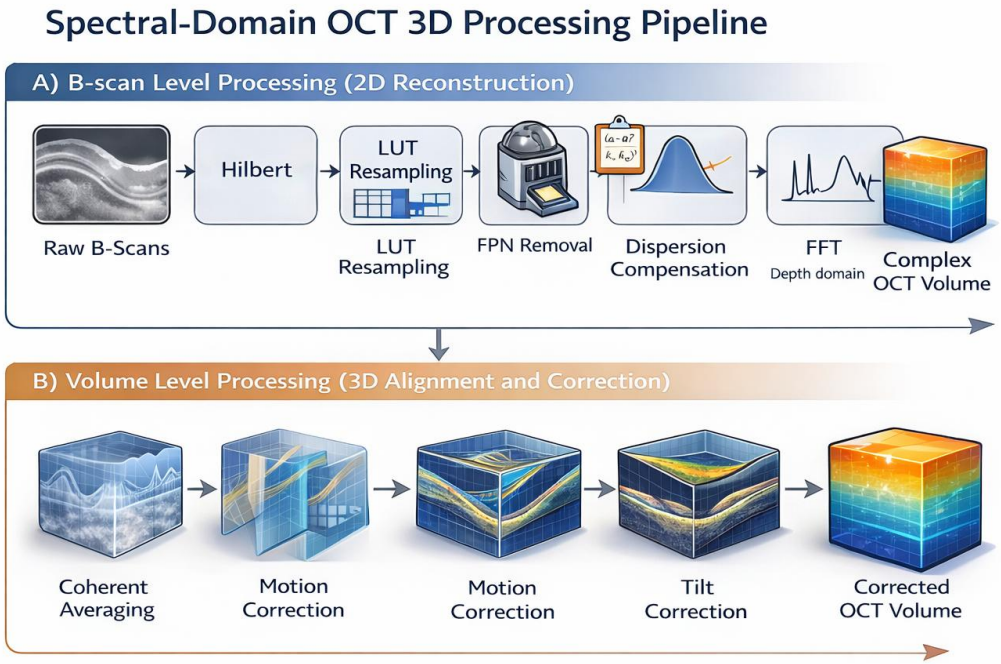
Why OCT reconstruction needs a processing pipeline

What we measure:

- OCT does not directly measure depth
- The detector records a spectral interferogram
- Signal is sampled:
  - uniformly in wavelength
  - non-uniformly in wavenumber (k)

Problems with raw data

- Non-uniform k-sampling
- Dispersion mismatch
- Fixed-pattern noise
- Motion and tilt



2D Processing (B-scan Level)

B-Scan Level Processing: Main Steps

Complex Signal Formation

Real signal → complex signal (real + imaginary)

01

02

k-Linearization

Non-uniform samples → uniform grid

Dispersion Compensation

Blurred reflector → sharp reflector

04

03

Noise Removal & Windowing

Noisy spectrum → smooth tapered spectrum

Complex Signal Formation

Reference selection:

- First acquired B-scan is selected as the reference frame

Complex signal formation:

- Raw spectral data is real-valued
- Hilbert transform is applied along the spectral dimension
- Produces a complex analytic spectrum

Why this is needed:

- Enables phase-based operations
- Required for dispersion compensation, and phase-sensitive motion correction.

k-Linearization (reSampling_LUT)

Inputs :

- rawData (complex matrix)
- rescaleParam (LUT)

Output:

- rescaledData

Description:

- FFT of each A-scan
- Zero-pad the FFT to a fixed length (2^{12})
- Inverse FFT to obtain an oversampled version of the spectrum
- Convert LUT positions to the padded grid
- Interpolate each A-scan at the LUT-defined positions

Fixed-Pattern Noise Removal and Spectral Windowing

Fixed-pattern noise (FPN):

- Let $S_{ref}(k,n)$ be the complex, k-linearized reference spectrum:

$$\begin{aligned}&\mathrm{FPN}(k)\\&= median_n\{\mathrm{R}[S_{\mathrm{ref}}(k,n)]\}\\&+\ i\, median_n\{\mathrm{I}[S_{\mathrm{ref}}(k,n)]\}\end{aligned}$$

- FPN-corrected spectrum:

Spectral windowing:

- Hann window applied along spectral dimension
- Reduces sidelobes and spectral leakage in FFT

Entropy-Based Dispersion Compensation

Dispersion phase model:

- Dispersion coefficients are estimated using the setDispCoeff function.

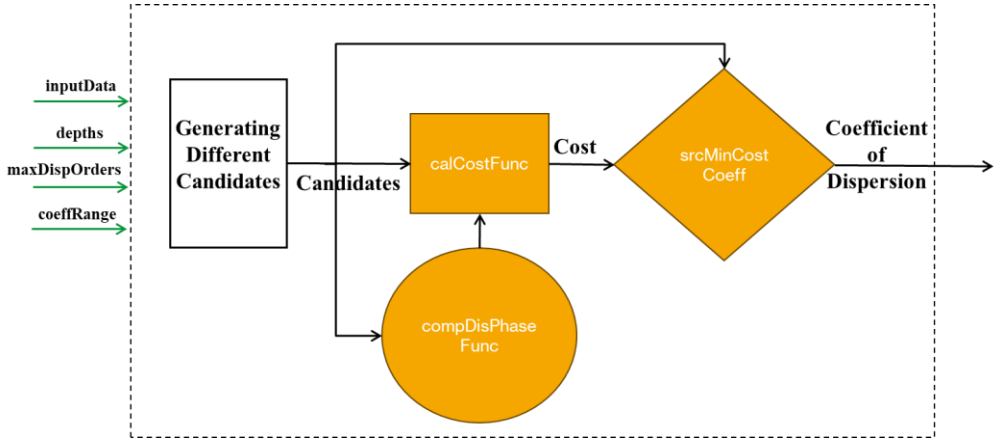
$$\phi_{\text{disp}}(k) = \sum_{n=2}^{N} a_n (k - k_0)^n$$

- a_n: dispersion coefficients
- k_0: reference wavenumber
- Lower-order (constant, linear) terms are omitted since they do not affect axial resolution

Phase compensation:

$$E_{\text{corrected}}(k) = E_{\text{raw}}(k)\, e^{-j\,\phi_{\text{disp}}(k)}$$

setDispCoeff



setDispCoeff

Inputs :

- rawData (complex matrix)
- depths (numeric vector)
- maxDispOrders (integer scalar)
- coeffRange (real scalar)

Output:

- dispCoeff

Description:

- Initialize coefficient vector
- Loop over dispersion orders (optimize one coefficient at a time)
- Evaluates the cost for each candidate
- Refinement by midpoint insertion near the minimum
- Select the coefficient with minimum cost

calCostFunc

Inputs :

- rawData (complex matrix)
- depths (numeric vector)
- maxDispOrders (integer scalar)
- arrCountDispCoeff (numeric vector)

Output:

- cost

Description:

- Apply dispersion compensation phase
- Reconstruct depth-domain OCT intensity
- Extract depth region of interest (ROI)
- Normalize ROI intensity to a probability distribution
- Compute Shannon entropy and sum to form the cost

Shannon entropy definition (discrete form):

$$H = -\sum_i p_i \log(p_i)$$

where p_i are the normalized intensities in normOCT.

compDisPhaseFunc

Inputs :

- rawData (complex matrix)
- maxDispOrders (integer scalar)
- arrCountDispCoeff (numeric vector)

Output:

- rescaledData (complex matrix)

Description:

- Read input dimensions
- Build a normalized k-axis
- Apply polynomial dispersion phase terms

srcMinCostCoeff

Inputs :

- arrCoeff (numeric vector)
- arrCost (numeric vector)

Output:

- frstMin
- scndMin

Description:

- Sort cost values
- Select the two best coefficients

Volume-Level Corrections

Apply Dispersion Compensation to Full Volume

Dispersion coefficients:

- Estimated once from a single representative B-scan
- Assumed constant across the volume

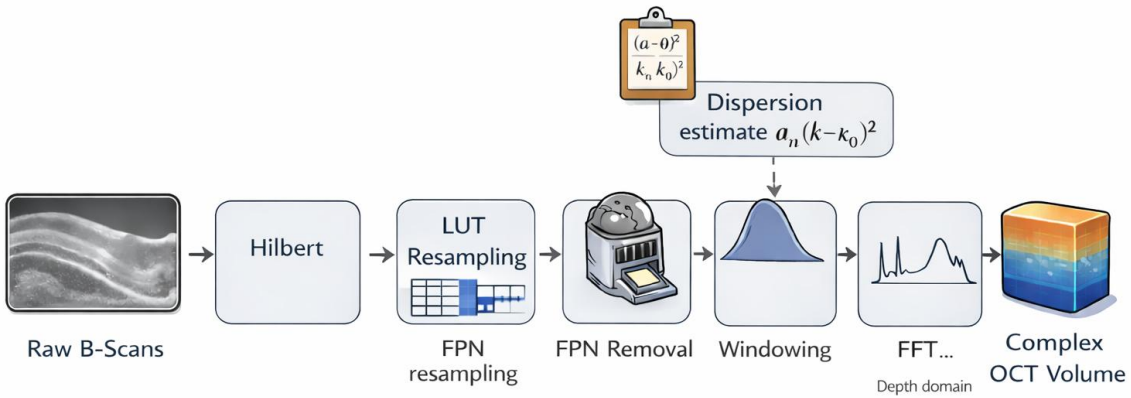
Loop over all B-scans

For each B-scan, the following steps are applied:

- Hilbert transform → complex spectral signal
- k-linearization using LUT
- Background / fixed-pattern noise subtraction
- Hann windowing
- Apply estimated dispersion coefficients
- FFT → depth-domain signal

Output

- Complex OCT volume is stored
- Both magnitude and phase are preserved



Phase-Corrected Coherent Averaging Across Volumes

Starting point:

- Multiple complex reconstructed OCT volumes
- First reconstructed volume is chosen as the phase reference

Phase alignment:

- Loop over remaining volumes
- Estimate bulk phase offset relative to the reference:

Let $S_{\text{ref}} = A\, e^{j\theta_{\text{ref}}}$ and $S_{\text{tar}} = A\, e^{j\theta_{\text{tar}}}$. Now multiply:

$$S_{\text{tar}} \cdot S_{\text{ref}}^{*} = B e^{j\theta_{\text{tar}}} \cdot A e^{-j\theta_{\text{ref}}} = AB\, e^{j\left(\theta_{\text{tar}} - \theta_{\text{ref}}\right)}$$

- Apply complex phase correction → Remove global phase drift between volumes

Coherent averaging:

- Phase-aligned volumes are averaged coherently
- Phase consistency maintained across the volume

Motion Correction (Axial and Lateral)

Reference selection

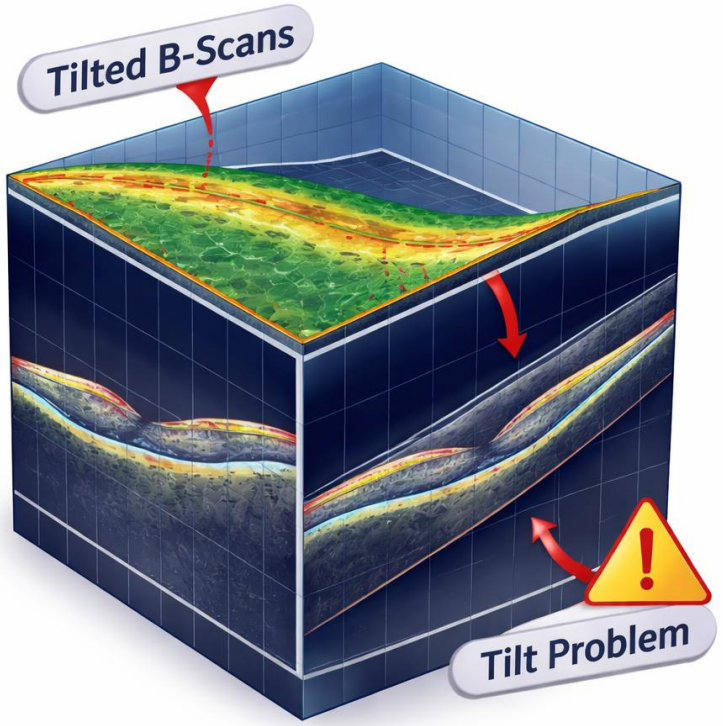
- The middle B-scan of the volume is selected as the reference frame

Shift estimation (frequency domain):

- For each B-scan in the volume compute Fourier transforms
- Estimate relative shift from the phase difference

Refinement:

- Apply the axial shift to align the B-scans using the circshift function



Tilt Correction (Fast-Scan Direction)

Reference selection

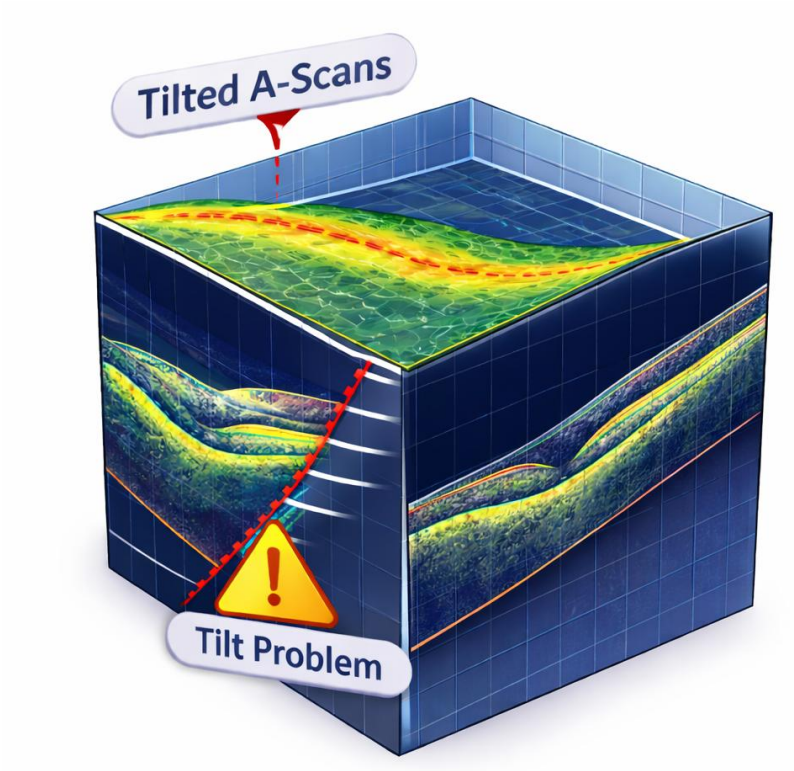
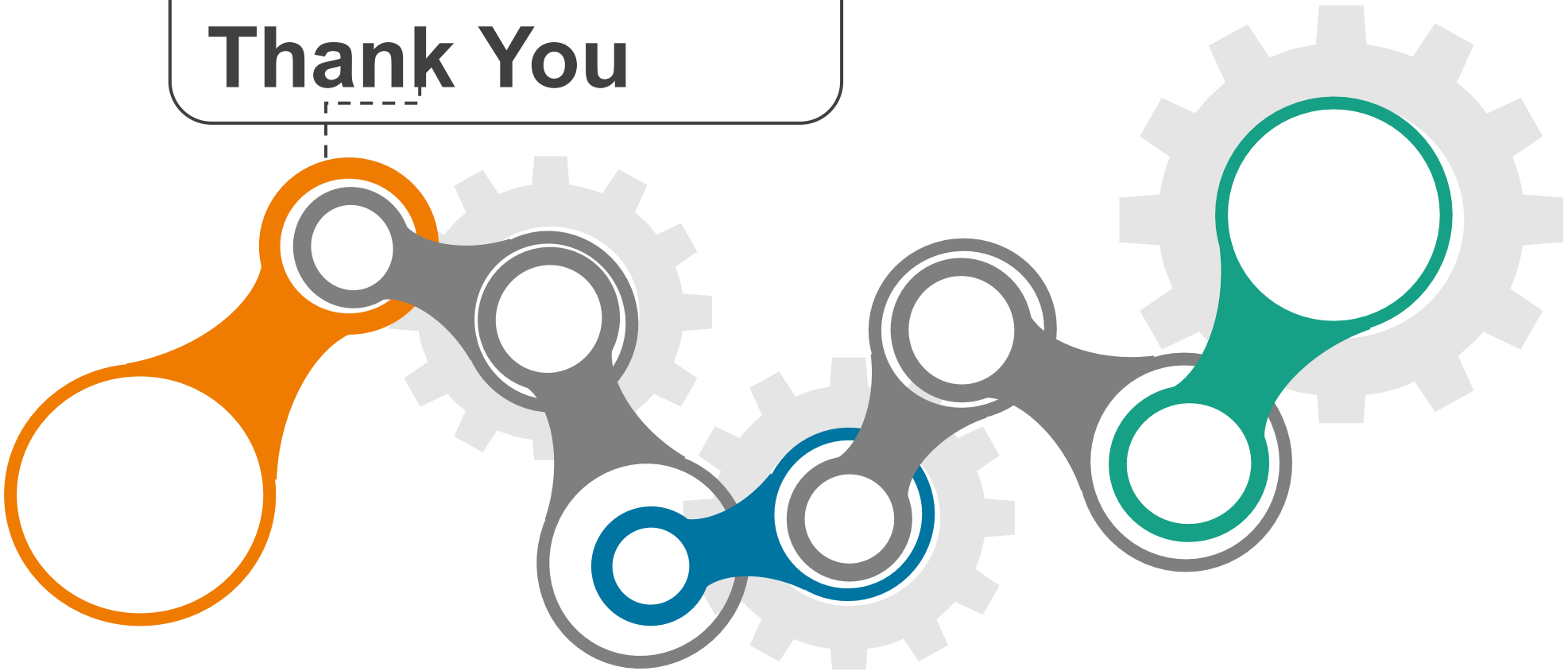
- Select the middle A-scan across the volume and use it Used as the reference profile

Shift estimation (frequency domain):

- For each A-scan along the volume, compute Fourier transforms
- Estimate relative shift from the phase difference

Refinement:

- Since, dftregistration_vol functions treat dimension 3 as the frame dimension, permute volume dimensions
- Apply the estimated axial and lateral shift to align the A-scans on permuted volume
- Permute dimensions back to original order



The End
Thank You